Lab 2:

1. Collection Management Using Arrays, Slices, and Maps

Write a Go program that:

(1) uses a slice to store a collection of elements (e.g., student names or integers) and implements add, remove-by-index, and update operations, printing the slice after each step;

(2) uses a map to store key–value pairs (e.g., subject → marks) and performs insert, delete, and lookup operations, printing the map after each operation.

(3) Push the source file to your GitHub repository. Upload screenshots of the output.

C: > Users > Jigyasa singh > Downloads > Lab2main.go > ...

```go
package main

import "fmt"

func main() {

    // Part 1: Slice Operations
    fmt.Println("----- SLICE OPERATIONS -----")

    students := []string{"Jigyasa Singh", "Anania", "Ravi"}
    fmt.Println("Original Slice:", students)

    // Add an element
    students = append(students, "Neha")
    fmt.Println("After Adding:", students)

    // Remove element by index
    index := 1
    students = append(students[:index], students[index+1:]...)
    fmt.Println("After Removing:", students)

    // Update an element
    students[1] = "Rohit"
    fmt.Println("After Updating:", students)

    // Part 2: Map Operations
    fmt.Println("\n----- MAP OPERATIONS -----")

    marks := map[string]int{
        "Maths":   85,
        "Science": 90,
    }

    fmt.Println("Original Map:", marks)

    // Insert a key-value pair
    marks["English"] = 88
    fmt.Println("After Inserting:", marks)
```

```go
// Delete a key-value pair
delete(marks, "Science")
fmt.Println("After Deleting:", marks)

// Lookup an element
value, exists := marks["Maths"]

if exists {
    fmt.Println("Maths Marks:", value)
} else {
    fmt.Println("Subject not found")
}

fmt.Println("Final Map:", marks)
```

OUTPUT:

```
PS C:\Users\Jigyasa singh> go run "c:\Users\Jigyasa singh\Downloads\Lab2Main.go
----- SLICE OPERATIONS -----
Original Slice: [Jigyasa Singh Anania Ravi]
After Adding: [Jigyasa Singh Anania Ravi Neha]
After Removing: [Jigyasa Singh Ravi Neha]
After Updating: [Jigyasa Singh Rohit Neha]

----- MAP OPERATIONS -----
Original Map: map[Maths:85 Science:90]
After Inserting: map[English:88 Maths:85 Science:90]
After Deleting: map[English:88 Maths:85]
Maths Marks: 85
Final Map: map[English:88 Maths:85]
PS C:\Users\Jigyasa singh>
```